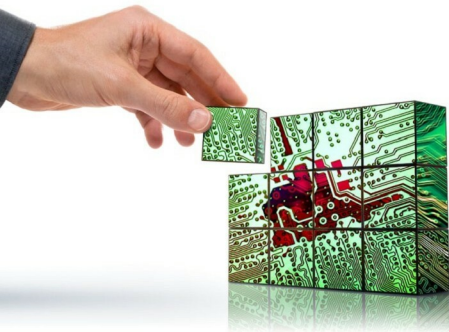


Embedded Electronics: In Nutshells and Capsules

Not figuratively but literally so, this article sheds light on how embedded electronics is moving from industrial and military equipment to everyday things—from nutshells and capsules, trees and towers to saucepans and soccer balls, making life easier and better in what we could now call the ‘Electronics of Things’ era



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Electronics has encircled our everyday life. The physical objects we see around us, phones, automobiles, bicycles or electrical appliances, are all equipped with smarter technology, which is increasing our dependency on them. In this era, we are enabling the objects around us to perform various tasks independently or by following a single command without any error. Also, they are integrated with each other smartly. This trend is evolving as we get to see results that were at some point of time beyond our imagination. All this would

not only ease our daily life with minimal usage of time but will also help us in a cost-efficient manner,” says Vivek Tyagi, country sales director, Freescale Semiconductor.

This is precisely what we have chosen to call the ‘Electronics of Things’—an age wherein embedded electronics has grown beyond elite objects into everyday ones, making them smarter and enabling them to do their and our jobs more efficiently.

Who would have imagined...?

A few years ago, who would have imagined electronics in coffee mugs, saucepans, innerwear, doctors’ pre-

scriptions or even the capsules you swallow? Interestingly, today there are applications that surprise even those who have been in the field of electronics all their life!

Capsules. “One particular application that I could have never fathomed is ‘eating’ electronics. Today, we have X-ray capsules that, when a person eats, go inside the body and create three-dimensional images of the intestinal walls and transmit those images to a watch, phone or any handheld gadget, which can then also be transmitted over the Internet to share the results with the doctor. Who would have imagined such a gadget even five years ago?” says Tyagi. You can read more about the first non-invasive, prep-free imaging test for colon cancer at www.check-cap.com.

Pregnancy belts. Priya Kuber, managing director, Arduino India, an open source project at the forefront of DIY innovation, exclaims, “Many such examples! Recently, I saw a belly belt which tweets every time a baby kicks inside a pregnant woman.” Whilst that is something like the Kickbee baby movement detector and tweeter, the Huggies Pregnancy Belt is a bit more advanced—a pair of belts worn by the mom- and dad-to-be, which emulates the baby’s kicks in the father’s belt every time the baby kicks inside the mother’s womb.

Prescription apps. “A health application on the prescription (Rx)—one that can be prescribed by a physician”—that is what excites Vipin N.,